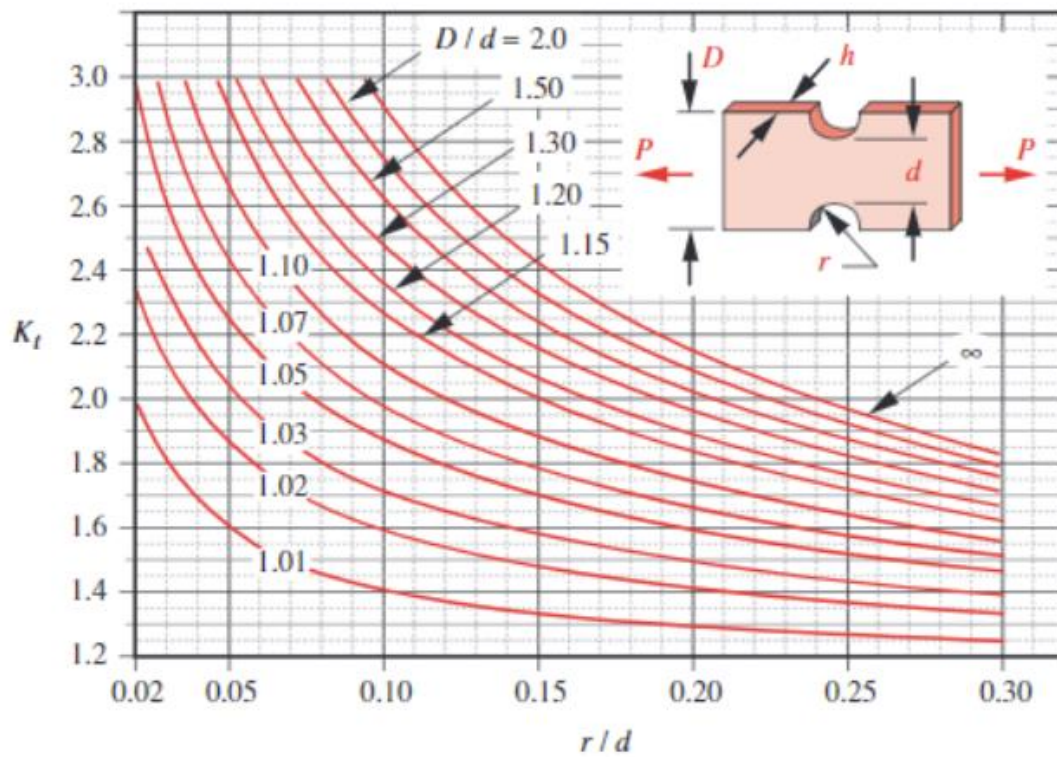


ME533 – Machine Design I

Fall 2023

Bailey Walke

C-11



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Introduction

Stress concentration factors are used to predict static failure around areas of concern. It is important to be able to determine the value of these factors to be able to account for failure in design. After years of experimentation and analysis, stress concentration factor curves and constants were derived to help predict these values. The purpose of this report is to further explore stress concentration and to try to validate these curves using Finite Element Analysis software, specifically the case of a flat notched bar in axial tension.

Background

Figure 1 approximates Stress-Concentration Factors for a notched flat bar in axial tension. These curves have been approximated from a series of experimental values. K_t varies with geometry. Constants A and b vary with the ratio of the bar thickness to the distance between the crest of the notches.

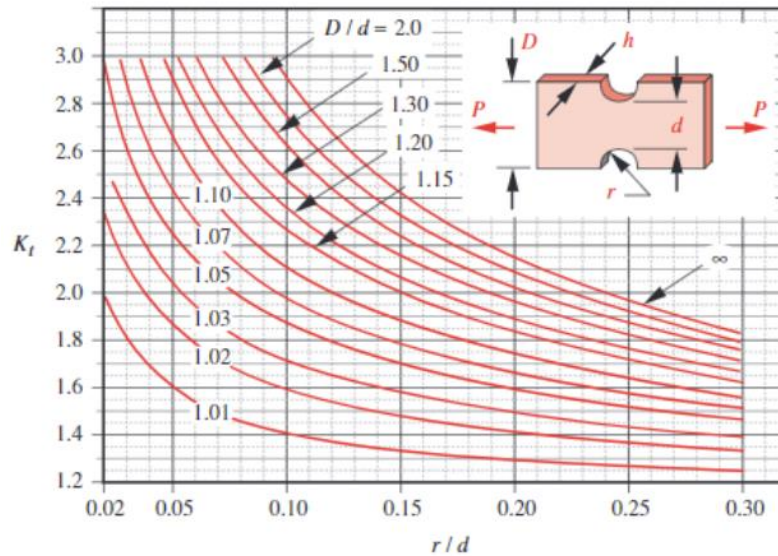


Figure 1 – Stress concentration factor curves for a notched bar in torsion

D/d	A	b
∞	1.109 50	-0.417 12
3.00	1.113 90	-0.409 23
2.00	1.133 90	-0.385 86
1.50	1.132 60	-0.365 92
1.30	1.158 60	-0.332 60
1.20	1.147 50	-0.315 07
1.15	1.095 20	-0.325 17
1.10	1.085 10	-0.299 97
1.07	1.091 20	-0.268 57
1.05	1.090 60	-0.241 63
1.03	1.051 80	-0.222 16
1.02	1.054 00	-0.188 79
1.01	1.042 60	-0.141 45

Figure 2 - Stress concentration factor calculation constants

$$K_t = A\left(\frac{r}{d}\right)^b$$

Equation 1 – Theoretical stress concentration factor

Using Solidworks Static simulation, two different geometric cases are modeled and tested; $D/d=1.1$ and $D/d=1.01$. In both cases, the width, distance ratio, and length are all held constant. Because the width does not affect the calculation of the geometric stress concentration factor, it is kept relatively small to speed up the computation time. Both samples have widths of 0.2 inches, and lengths of 10 times the largest notch radius on each side of the notch that is tested.

A total of 35 models will be tested; 14 for $D/d=1.01$ and 11 for $D/d=1.1$. The radii of each model are specified in Table 3 and Table 6. Each model will be tested under the same loading conditions and the maximum stress of the model will be recorded. Since the distance between the notches does and the thickness of the models do not change, the nominal stress for all the models should be the same. Therefore, this experiment is simplified by having one independent variable, notch radius, and one dependent variable, maximum stress.

Technical Approach

Procedure

Materials and Loading

By definition, the geometric stress concentration factor is only affected by the specimen's geometry.

Material factor and load do not change the results. However, Solidworks simulation software only operates within the elastic range and does not recognize phenomena such as strain hardening, meaning once the yield strength is reached, the simulation will continue along the linear path of the stress/strain curve. To ensure the most accurate results, the load must be well within the elastic limit of the material selected. For this experiment, alloy steel is used with a yield strength of 90ksi. For simplicity of calculation, each face of the model is subjected to 1lbf of axial force. This load will not exceed the yield strength of the material.

Split Line

A split line was modelled down the center of the specimen, as shown by Figure 3. This was used to apply mesh control to the stress concentration area and to probe for the maximum stress.



Figure 3 - Split line of model

Constraints

The simulation is similar to a tensile test, so the model must be constrained as such. The bottom right corner of the model is fixed in all directions, the bottom right is constrained in the y and z direction, and the top left corner is constrained in the z direction. This allows for the stress concentration area to deform as it would in a tensile test.

Mesh

Mesh convergence was completed for the largest and smallest radius of each simulation set. Mesh control was applied to the split line of the notched, and halved with each iteration of the convergence test. This was repeated until there was less than 1% relative error for the maximum stress of the model. The finest mesh between the two models was used for the whole data set to ensure mesh convergence for all models. While there is a very fine mesh in the stress concentration area, the rest of the model is left with a coarse mesh to speed up the computation time.

Calculations

The specimen illustrated in Case C-11 is a flat notched bar in axial tension. For simplicity purposes, a simple x-y-z cartesian coordinate system is used, and the load is applied in the positive/negative x-direction. Therefore, the stress component of concern is in the pure x-direction. To calculate the nominal stress, Equation 2 was used.

$$\sigma_{xx|nom} = \frac{P}{net\ ligament\ area}$$

Equation 1 – Nominal stress

$$K_t = \frac{\sigma_{xx|max}}{\sigma_{xx|nom}}$$

Equation 2 – FEM stress concentration factor calculation

All calculations for both the FEM and theoretical stress concentration factors were done using Excel. The tabulation tables for both cases are included below.

$$D/d = 1.1$$

Finite Element Model

Length: 60 in

Width: 11 in

Thickness: 0.2 in

Distance between notches: 10in

Force applied: 1lbf (each side)

- Element type



Figure 4 - Model 1.1-1

- Boundary and loading conditions



Figure 5 - Model 1.1-1 loading conditions

Mesh Convergence

- 1.1-1: $r/d=0.05$

- Element size: 1.5 in
- Mesh control size: 0.0625 in
- Maximum of stress component of concern: 1.396 psi
- Relative error: 0.72%

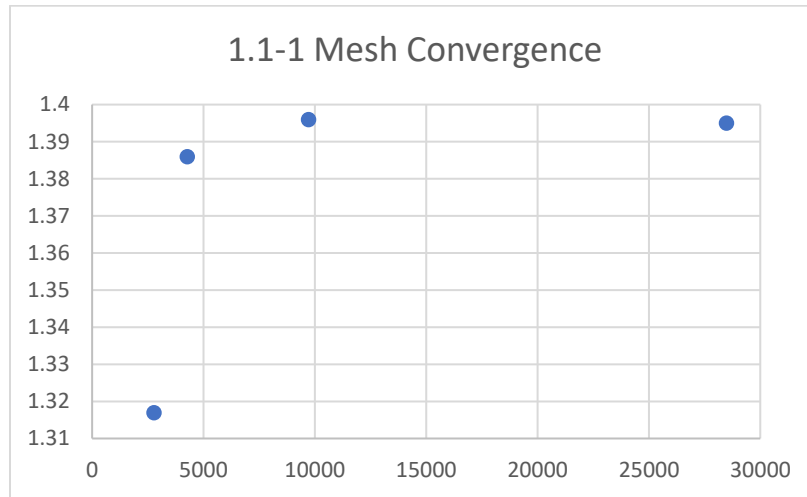


Figure 6 - Model 1.1-1 mesh convergence

1.1-1				
Standard Mesh Size	Mesh Control Size	# of Elements	Max Stress	Relative Error
1.5	0.25	2771	1.317	
1.5	0.125	4267	1.386	4.978354978
1.5	0.0625	9708	1.396	0.716332378
1.5	0.03125	28474	1.395	-0.071684588

Table 1 - Model 1.1-1 mesh convergence

1.1-11: $r/d=0.3$

- Element size: 1.5 in
- Mesh control size: 0.0625 in
- Maximum of stress component of concern: 0.81 psi
- Relative error: 0.12%

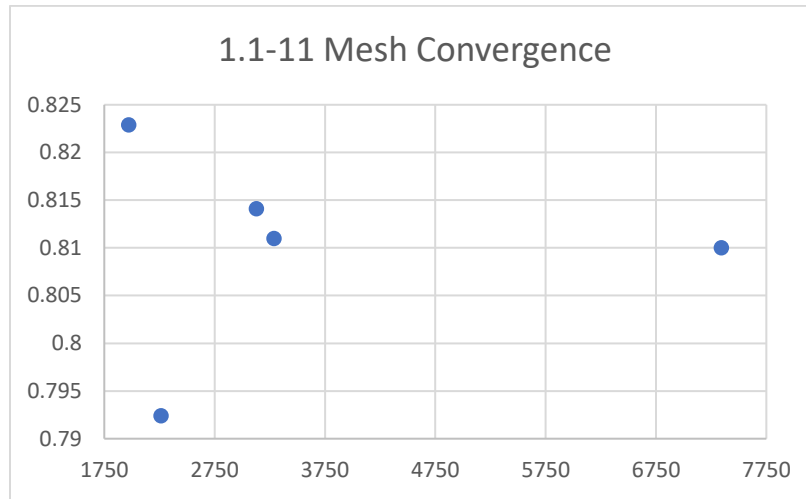


Figure 7 - Model 1.1-11 mesh convergence

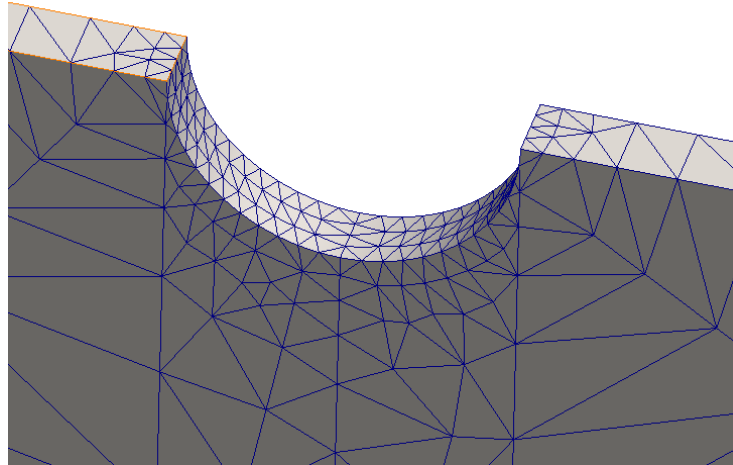
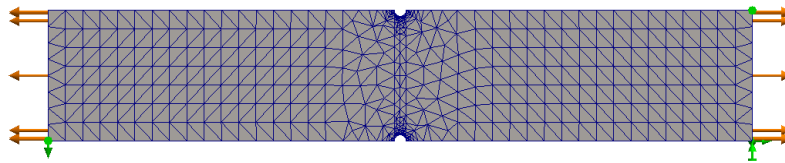
1.1-14				
Standard Mesh Size	Mesh Control Size	# of Elements	Max Stress	Relative Error
1.5	1	1972	0.8229	
1.5	0.5	2264	0.7924	-3.70640418
1.5	0.25	3127	0.8141	2.738515901
1.5	0.125	3288	0.811	-0.380788601
1.5	0.0625	7343	0.81	-0.123304562

Table 2 - Model 1.1-11 mesh convergence

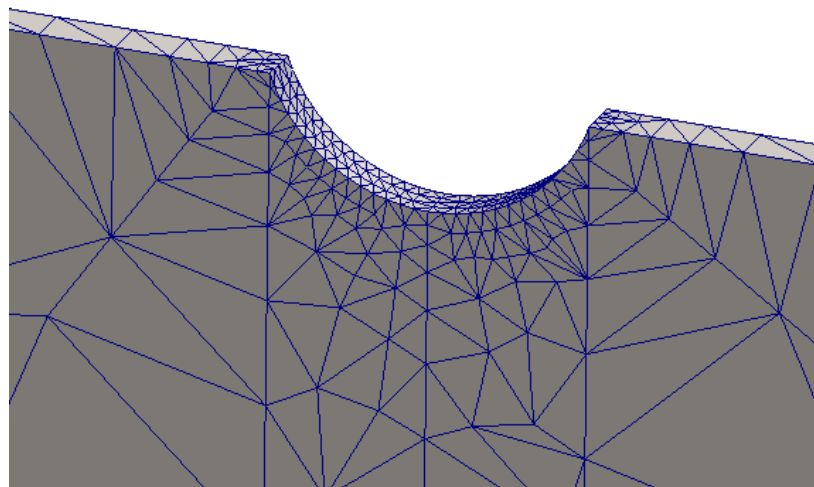
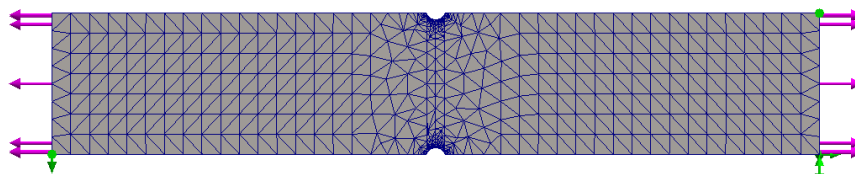
0.0625-inch mesh control was used on all models, since the stress converged for the two most extreme cases of the radii.

Mesh

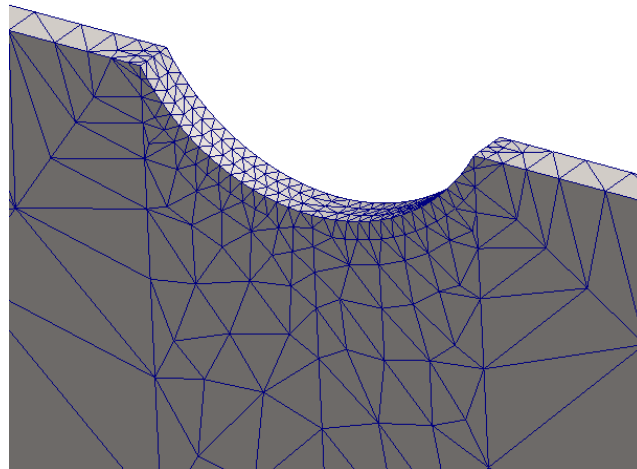
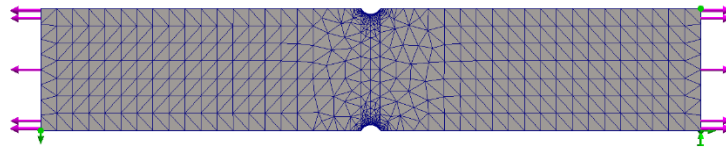
- Mesh of stress concentration area (all points)
 - o $r=0.5$



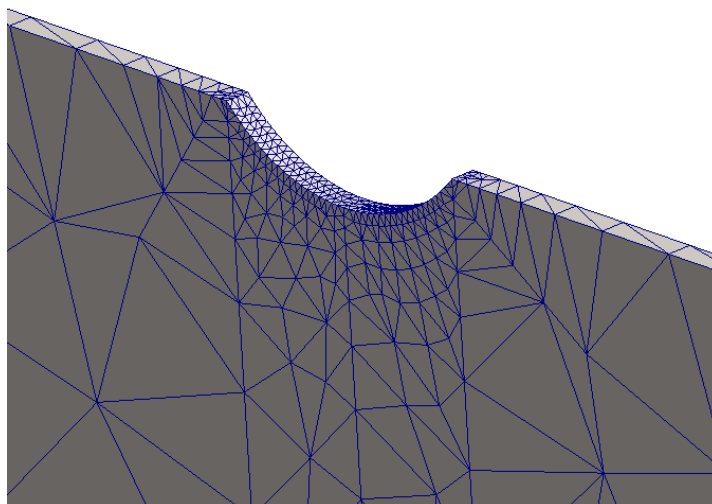
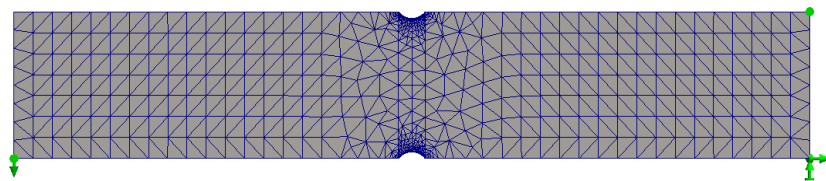
- $r=0.75$



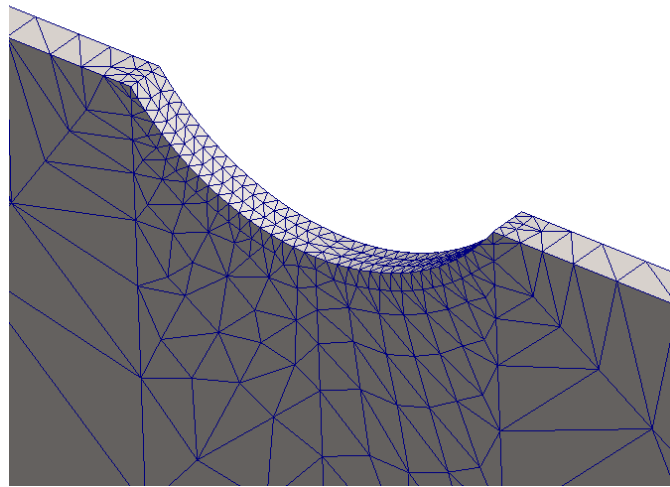
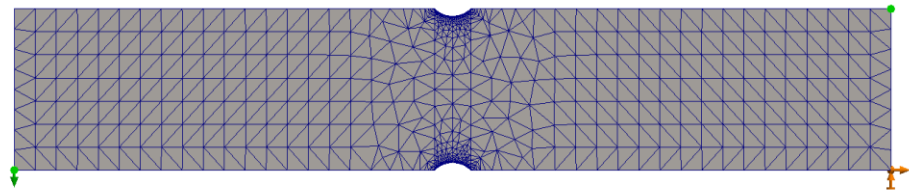
- $r=1.0$



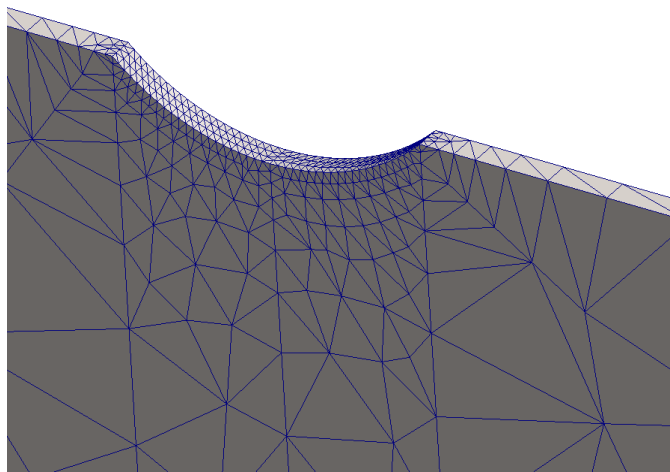
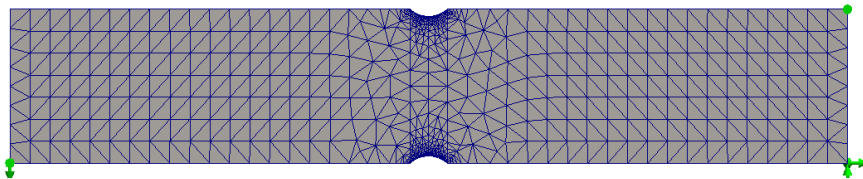
- $r=1.25$



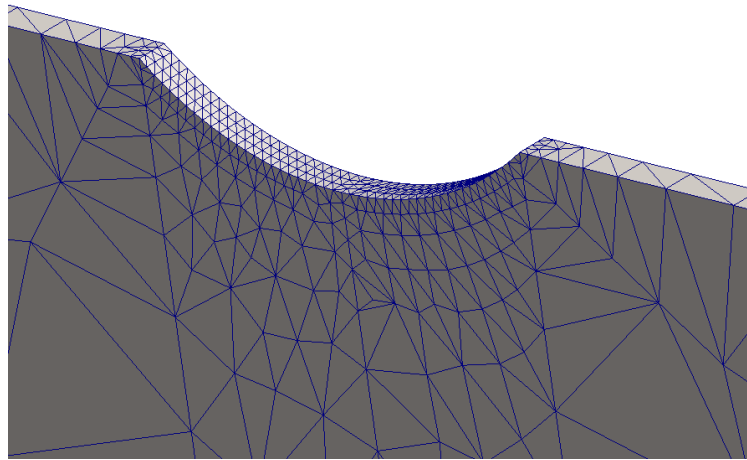
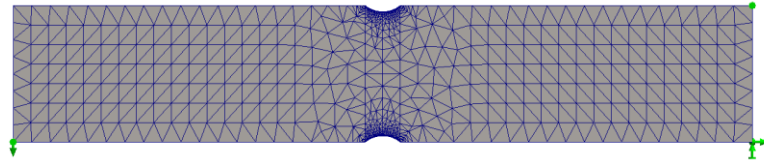
- $r=1.5$



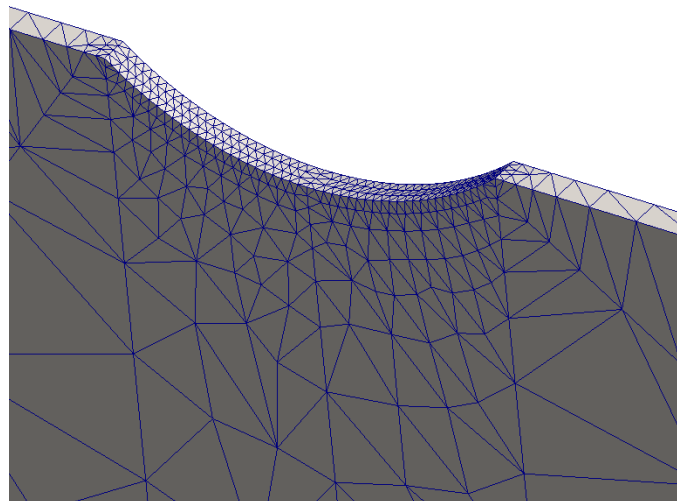
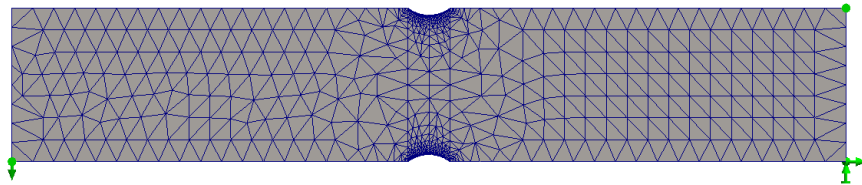
- $r=1.75$



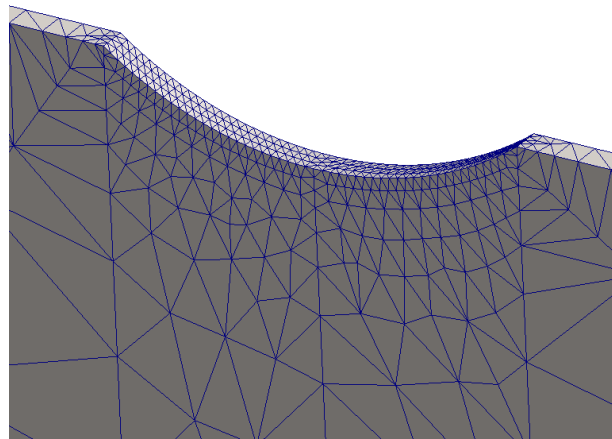
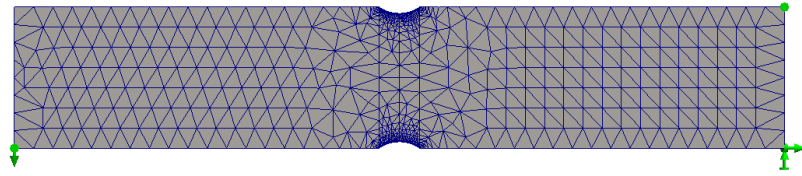
- $r=2.0$



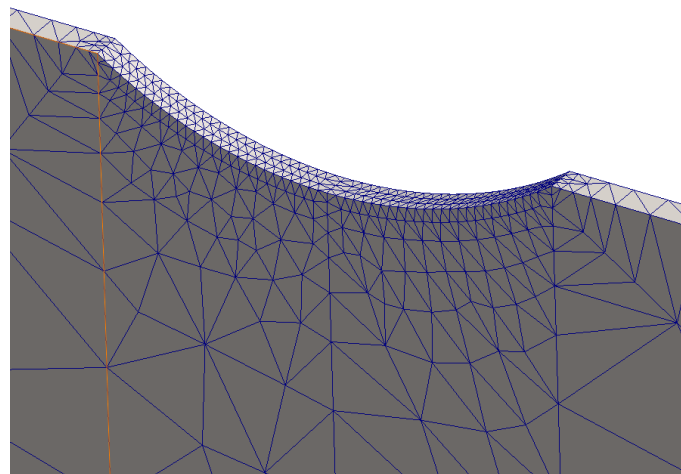
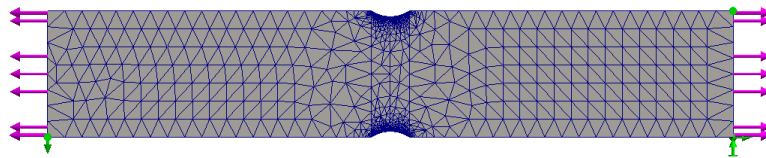
- $r=2.25$



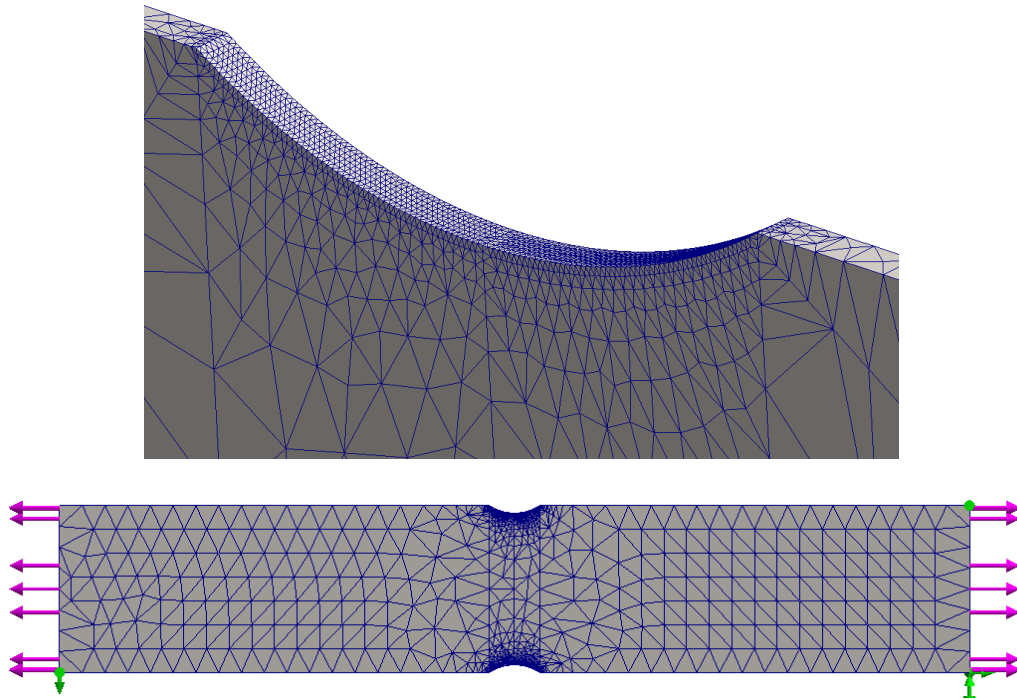
-r=2.5



- r=2.5



- $r=2.75$



Results

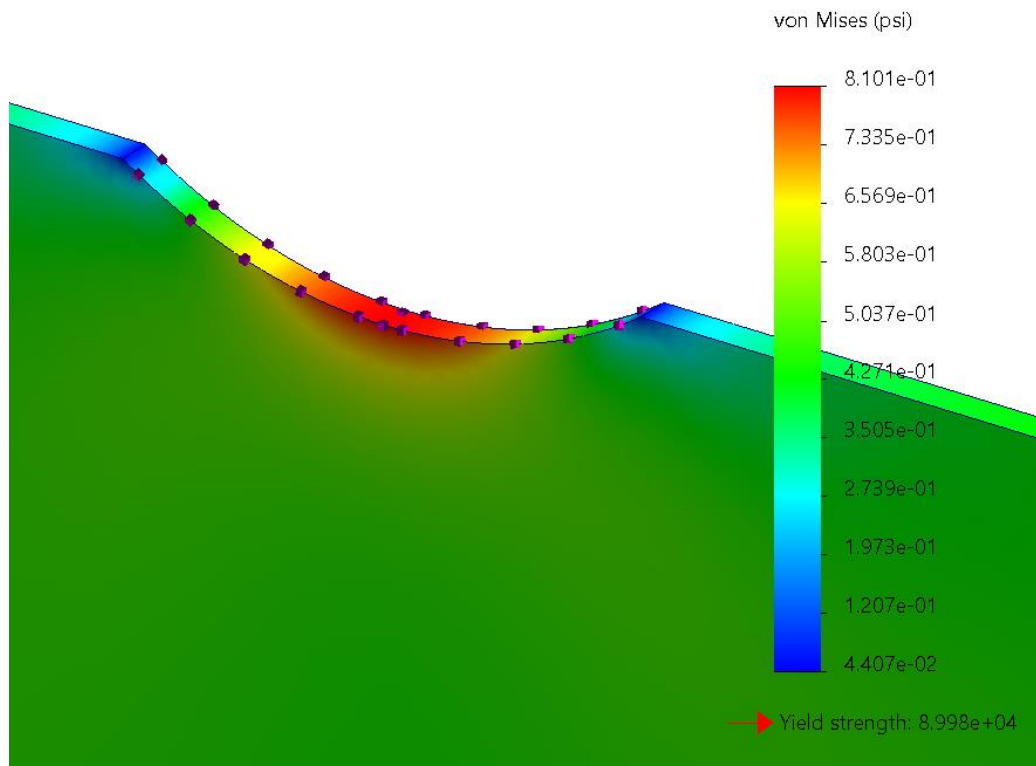


Figure 8 - Model 1.1-1 Von-mises stress contour plot

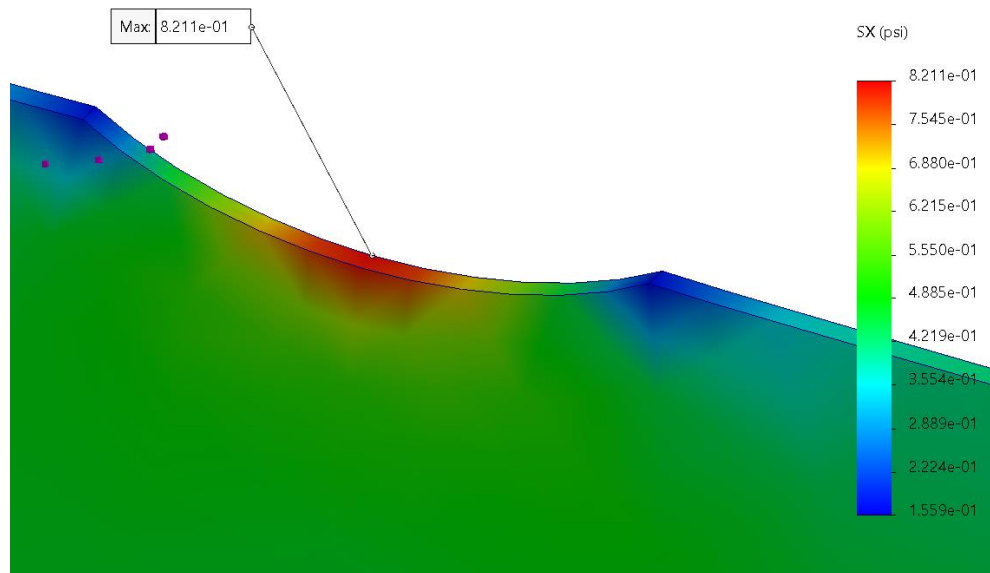


Figure 9 - Model 1.1-1 deformed nominal stress contour plot

Tabulation

Model #	D	d	D/d	L	h	P (lbf)	r	r/d	FEM				Theoretical			% Difference (FEM to Theoretical)
									net ligament area	nominal stress	maximum stress	K _t	A	b	K _t	
1.1-1	11	10	1.1	60	0.2	1.1	0.5	0.05	2	0.5	1.387	2.774	1.0851	-0.29997	2.665261	-4.07986625
1.1-2							0.7	0.075			1.208	2.416			2.360031	-2.371553463
1.1-3							1	0.1			1.096	2.192			2.16491	-1.251341581
1.1-4							1.2	0.125			1.023	2.046			2.024742	-1.049917318
1.1-5							1.5	0.15			0.9704	1.9408			1.91698	-1.242559279
1.1-6							1.7	0.175			0.9298	1.8596			1.830356	-1.597704942
1.1-7							2	0.2			0.8972	1.7944			1.75849	-2.042118138
1.1-8							2.2	0.225			0.8694	1.7388			1.697444	-2.436341191
1.1-9							2.5	0.25			0.8471	1.6942			1.644636	-3.013698121
1.1-10							2.7	0.275			0.8269	1.6538			1.598281	-3.473669853
1.1-11							3	0.3			0.8102	1.6204			1.557104	-4.064962918

Table 3 - D/d=1.1 calculations

Stress Concentration Plot

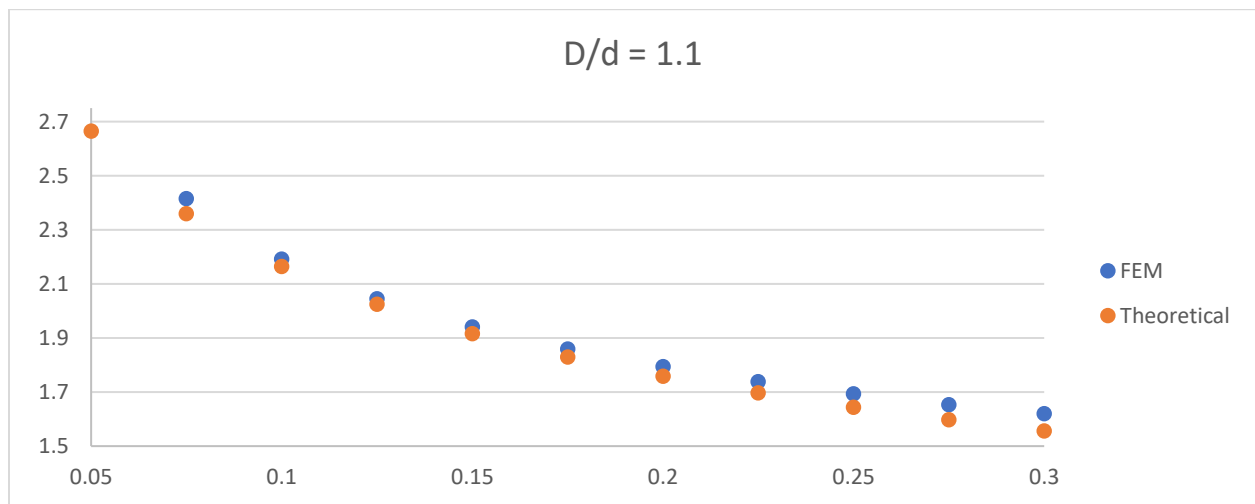


Figure 10 - D/d=1.1 FEM vs Theoretical plot

$$D/d = 1.01$$

Finite Element Model

Length: 60 in

Width: 10.1 in

Thickness: 0.2 in

Distance between notches: 10in

Force applied: 1lbf (each side)

- Element type

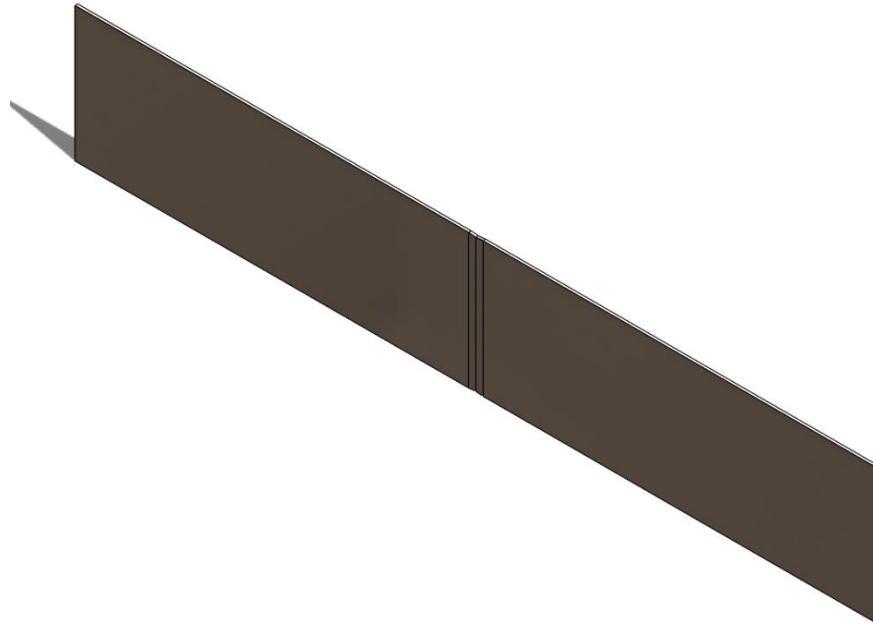


Figure 11 - Model 1.01-1

- Boundary and loading conditions



Figure 12 - Model 1.01-1 loading and boundary conditions

Mesh Convergence

- **1.01-1: $r/d=0.02$**
 - Element size: 1.5 in
 - Mesh control size: 0.03125 in

- Maximum of stress component of concern: 0.882 psi
- Relative error: 0.238%

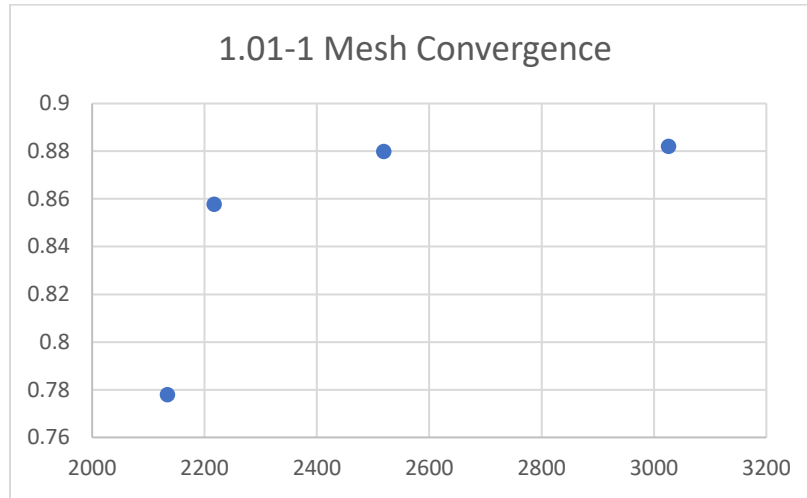


Figure 13 - Model 1.01-1 mesh convergence

1.01-1				
Standard Mesh Size	Mesh Control Size	# of Elements	Max Stress	Relative Error
1.5	0.25	2134	0.778	
1.5	0.125	2217	0.8578	9.302867801
1.5	0.0625	2519	0.8799	2.511649051
1.5	0.03125	3026	0.882	0.238095238

Table 4 - Model 1.01-1 mesh convergence

- **1.01-14: r/d=0.3**
 - Element size: 1.5 in
 - Mesh control size: 0.03125 in
 - Maximum of stress component of concern: 0.6137 psi
 - Relative error: 0.0325%

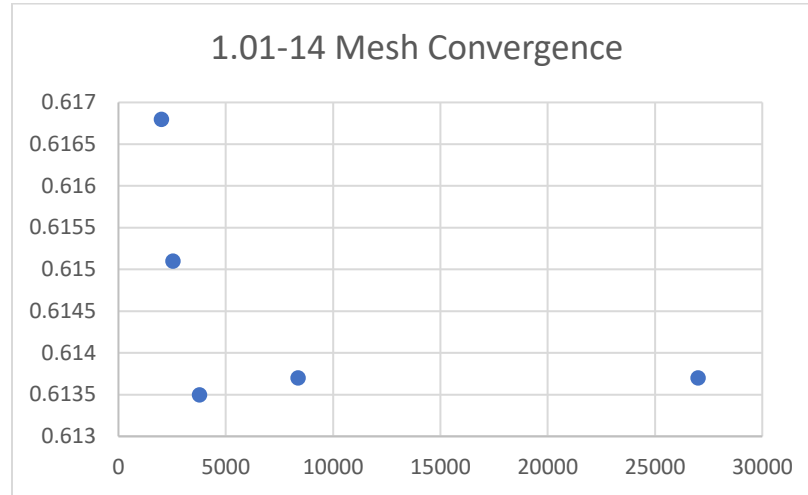


Figure 14 - Model 1.01-14 mesh convergence

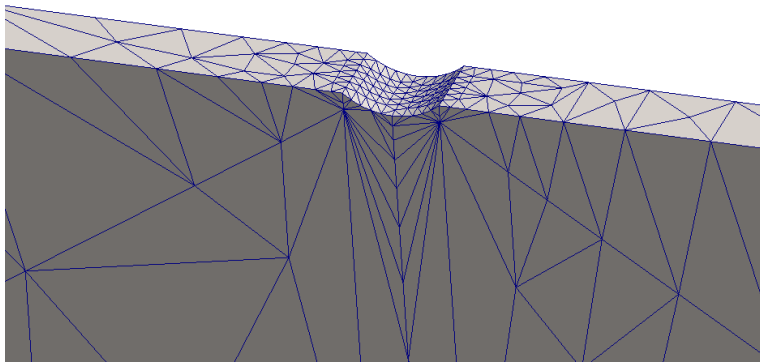
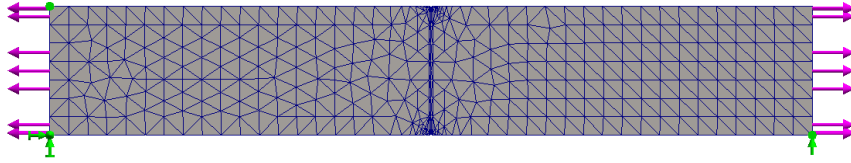
1.01-14				
Standard Mesh Size	Mesh Control Size	# of Elements	Max Stress	Relative Error
1	0.25	2006	0.6168	
1.5	0.125	2529	0.6151	-0.275616083
1.5	0.0625	3772	0.6135	-0.260120306
1.5	0.03125	8373	0.6137	0.032599837
1.5	0.015625	27010	0.6137	0

Table 5 - Model 1.01-14 mesh convergence

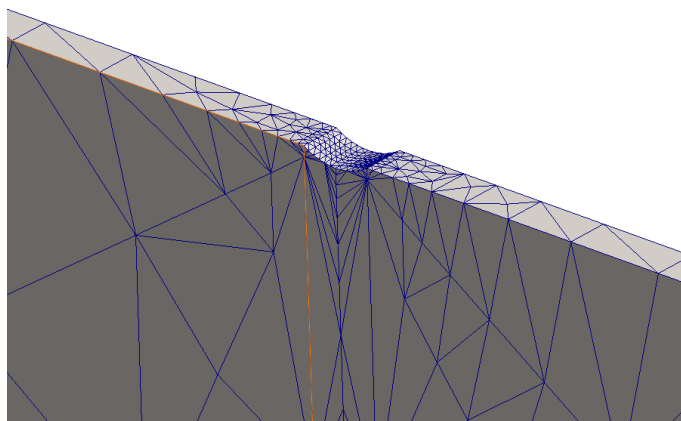
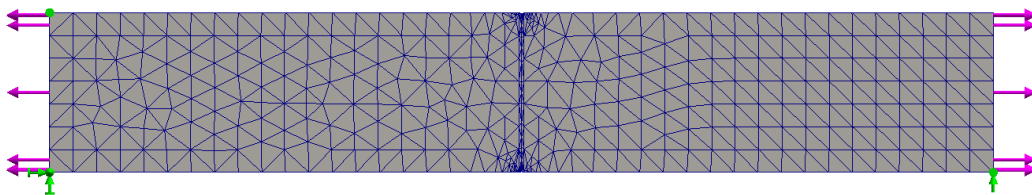
0.03125 inch mesh control was used on all models, since the stress converged for the two most extreme cases of the radii.

Mesh

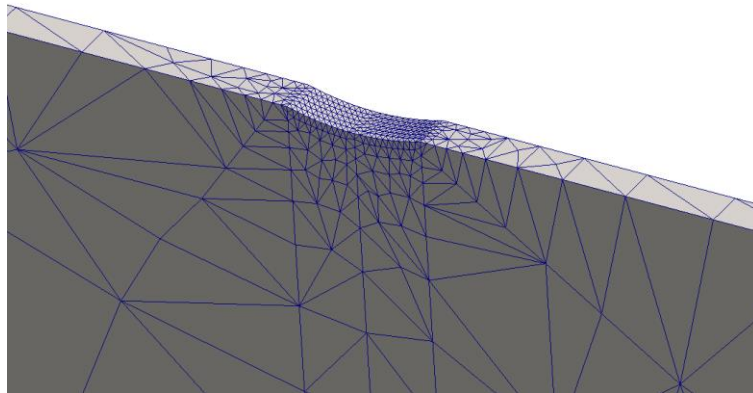
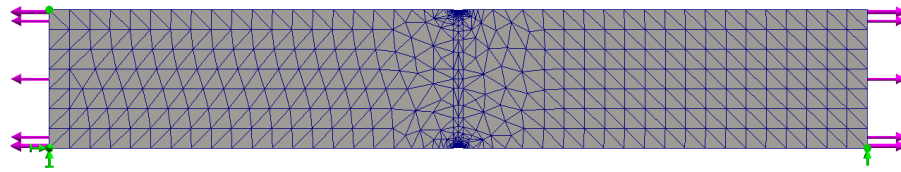
- $r=0.2$



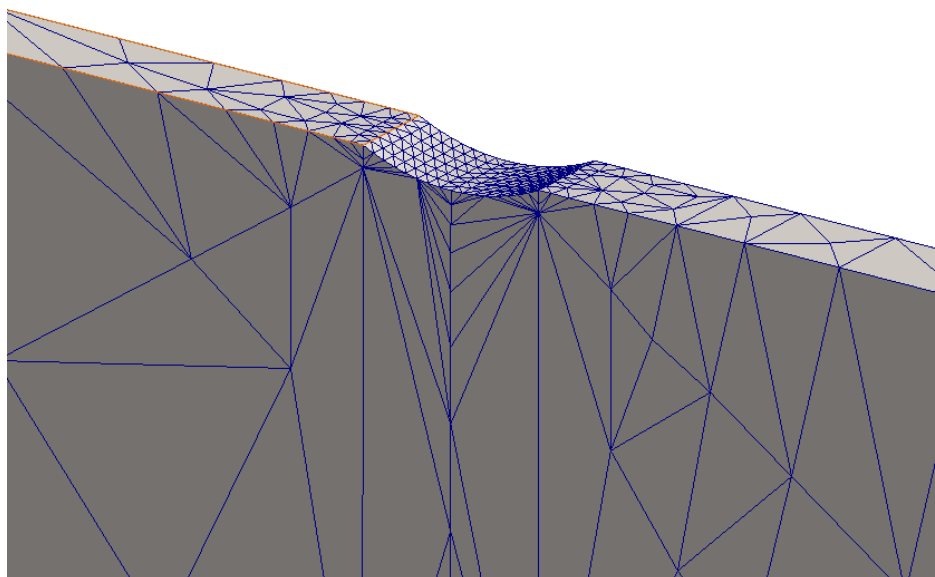
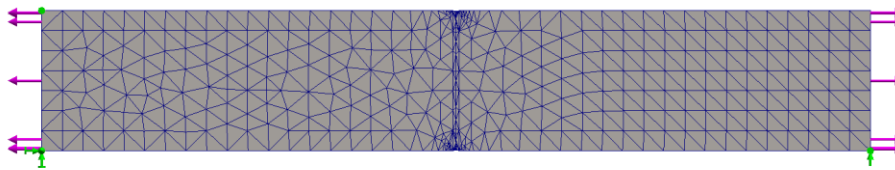
- $r=0.3$



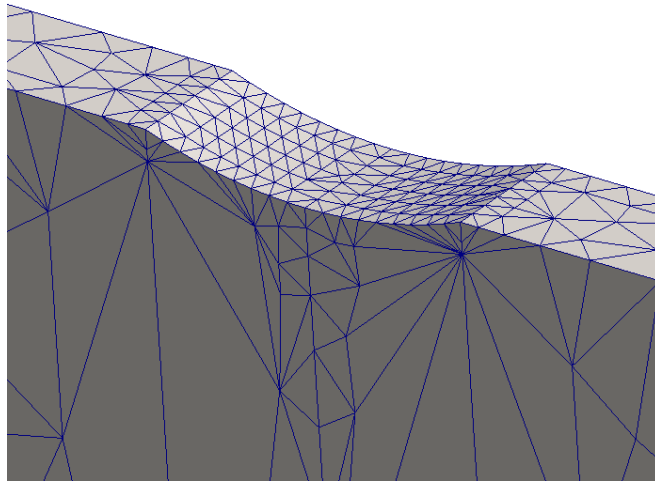
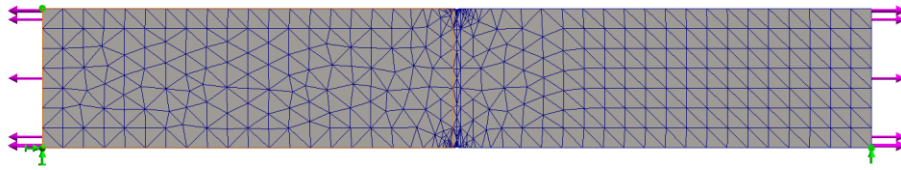
- $r=0.4$



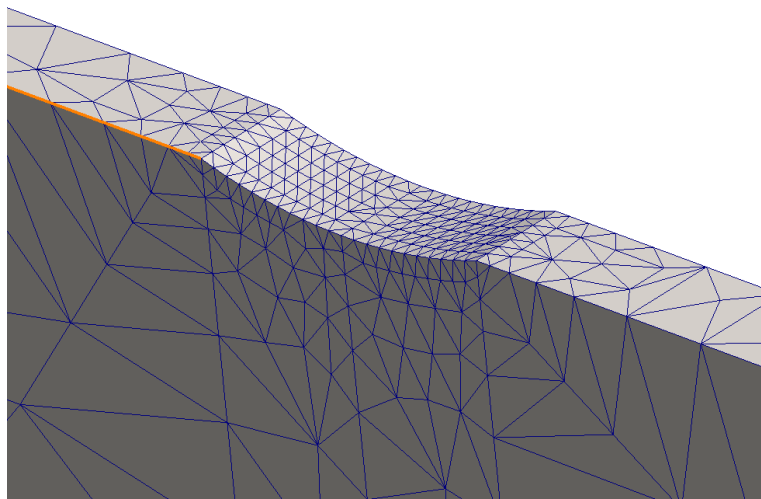
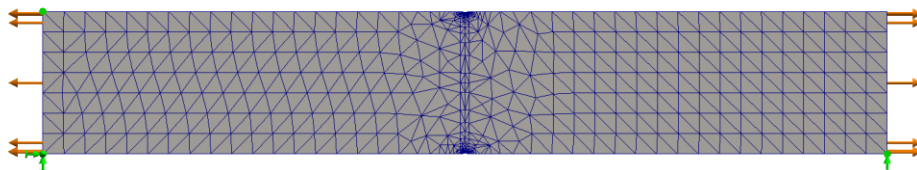
- $r=0.5$



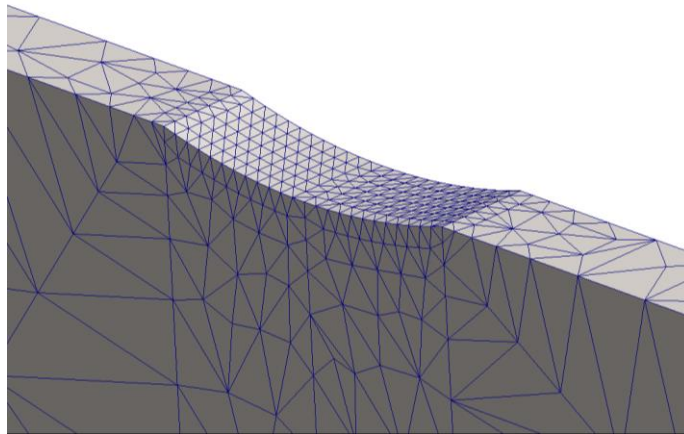
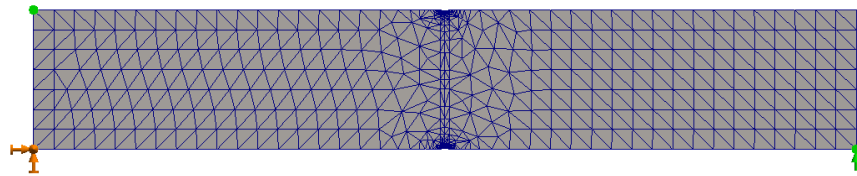
- $r=0.75$



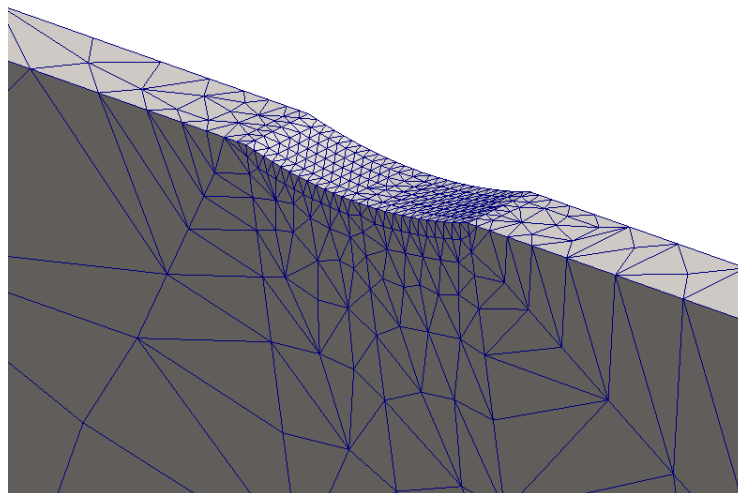
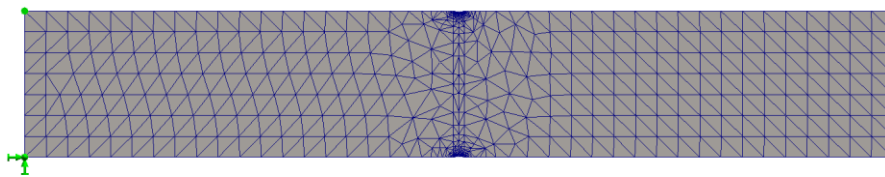
- $r=1.0$



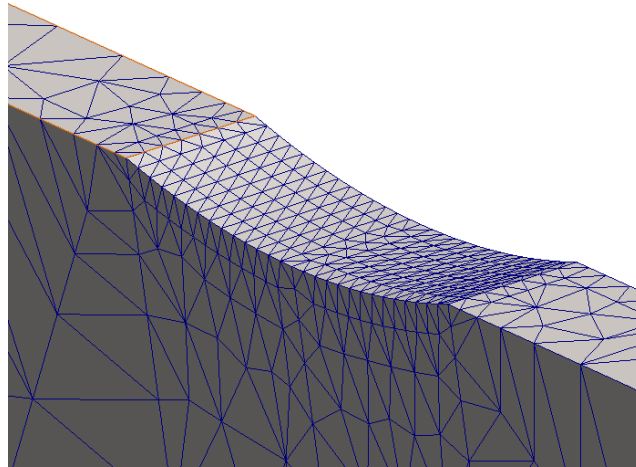
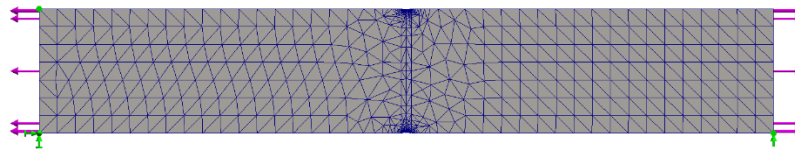
- $r=1.25$



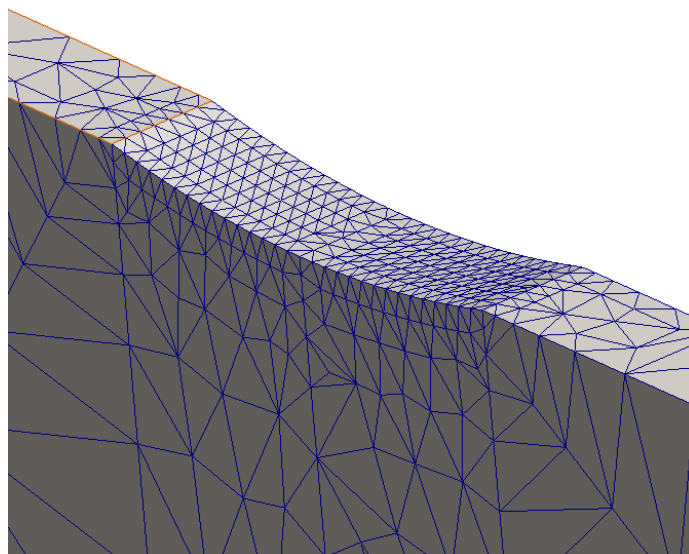
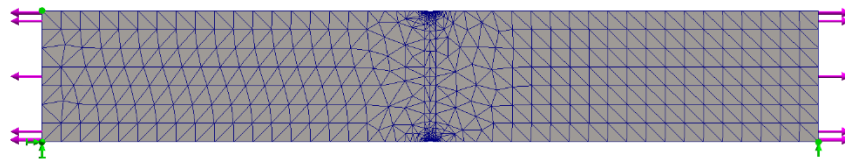
- $r=1.5$



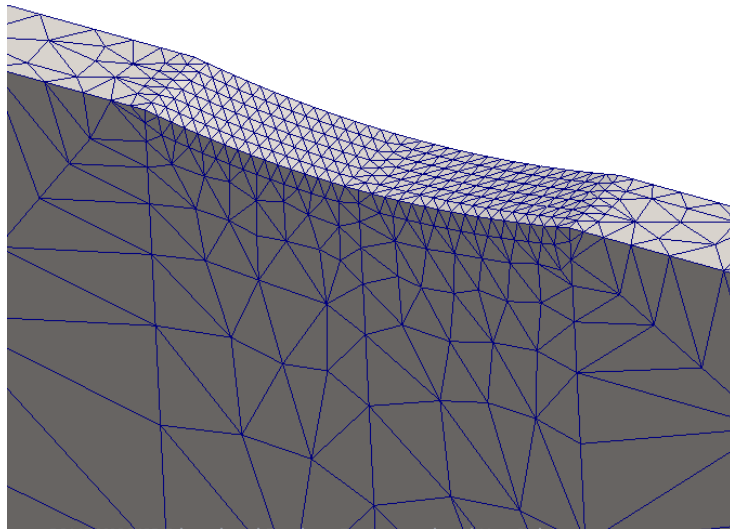
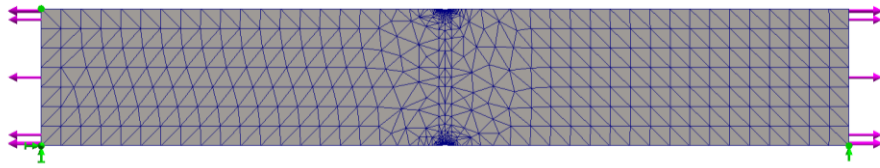
- $r=1.75$



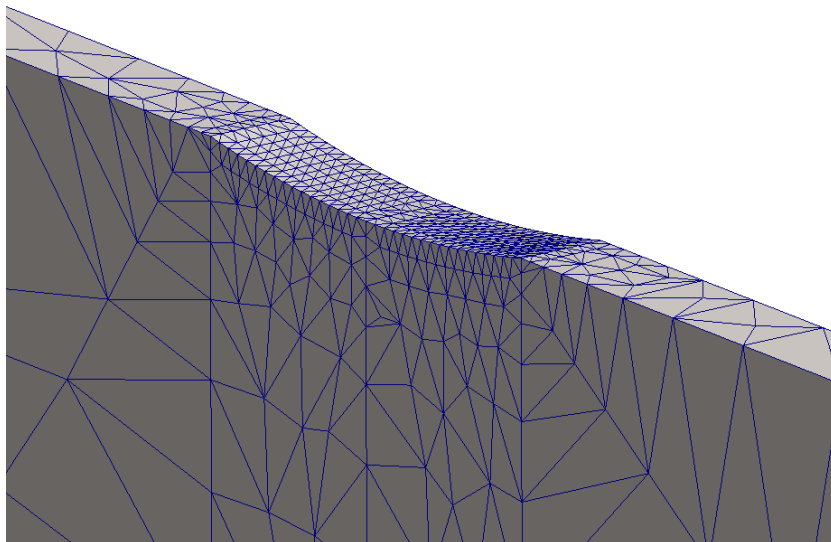
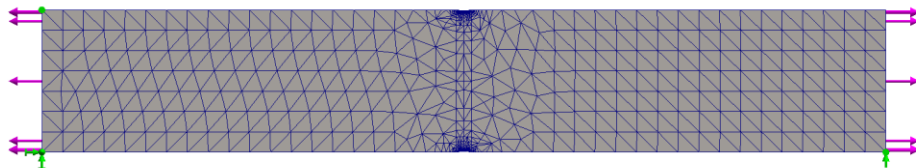
- $r=2.0$



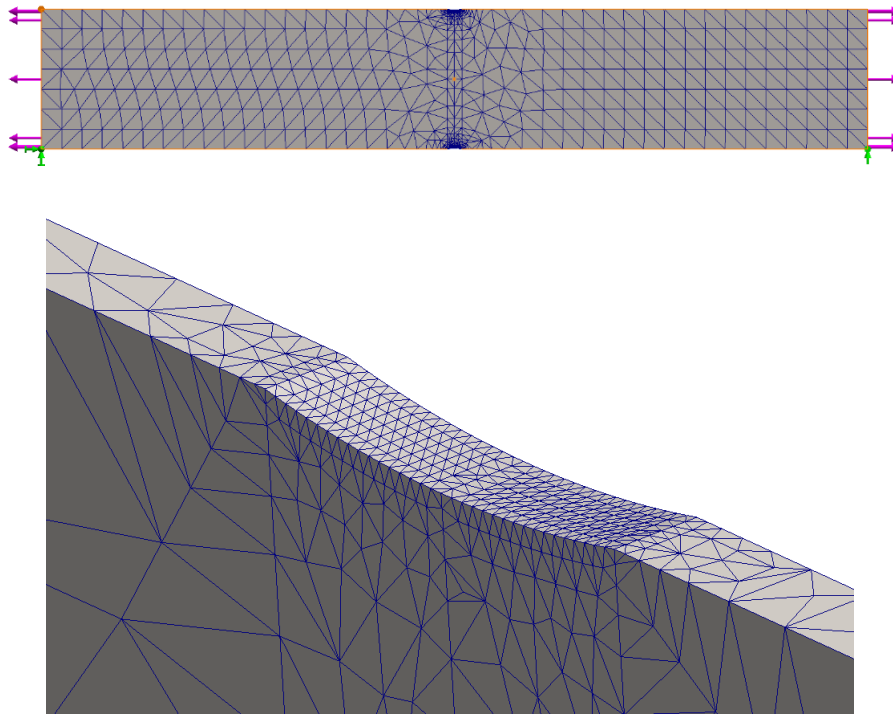
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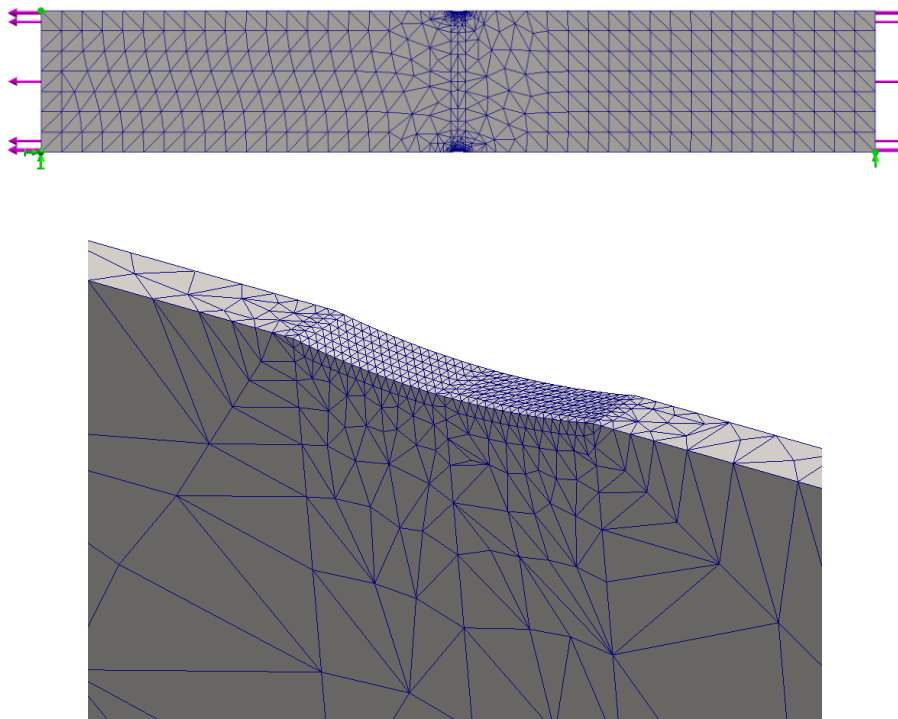
- $r=2.5$



- $r=2.75$



- $r=3.0$



Results

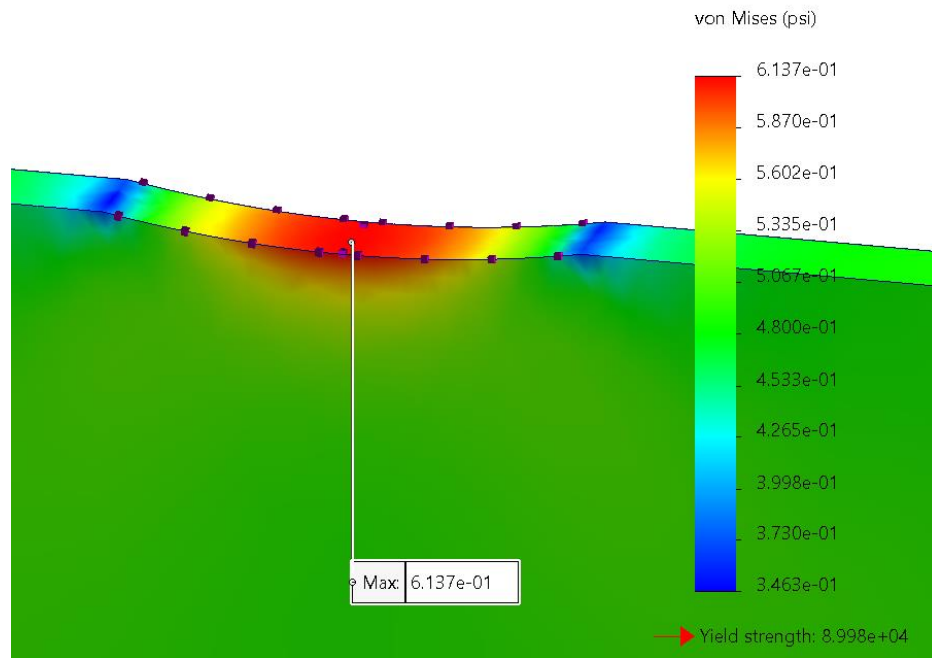


Figure 15 - Model 1.01-1 Von-mises contour plot

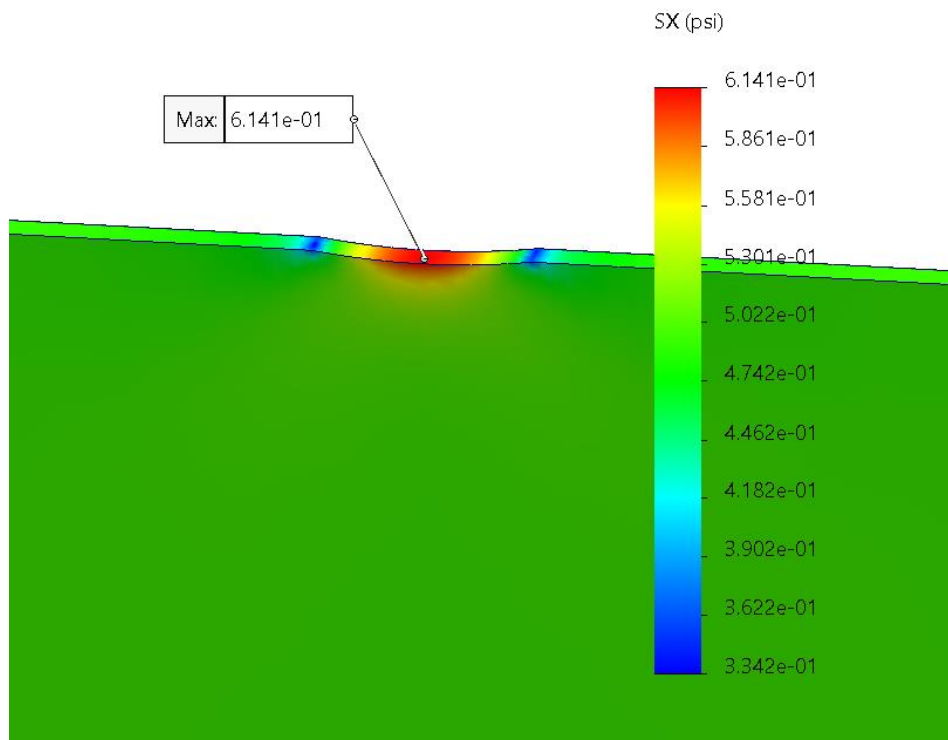


Figure 16 - Model 1.01-1 deformed nominal stress contour plot

Tabulation

									FEM				Theoretical			% Difference (FEM to Theoretical)
Model #	D	d	D/d	L	h	P (lbf)	r	r/d	net ligament area	nominal stress	maximum stress	K _t	A	b	K _t	
1.01-1	10.1	10	1.01	60	0.2	1	0.2	0.02	2	0.5	0.8641	1.7282	1.0426	-0.14145	1.813164	4.685939136
1.01-2							0.3	0.03			0.8359	1.6718			1.712099	2.35378767
1.01-3							0.4	0.04			0.7825	1.565			1.643828	4.795383697
1.01-4							0.5	0.05			0.7604	1.5208			1.592753	4.517520041
1.01-5							0.75	0.075			0.716	1.432			1.503974	4.785582884
1.01-6							1	0.1			0.7075	1.415			1.444002	2.008429239
1.01-7							1.25	0.125			0.6833	1.3666			1.399136	2.325405449
1.01-8							1.5	0.15			0.6659	1.3318			1.363514	2.325904415
1.01-9							1.75	0.175			0.6522	1.3044			1.334105	2.226581133
1.01-10							2	0.2			0.6418	1.2836			1.309143	1.951110162
1.01-11							2.25	0.225			0.6329	1.2658			1.287513	1.686402546
1.01-12							2.5	0.25			0.6256	1.2512			1.268467	1.361234254
1.01-13							2.75	0.275			0.6194	1.2388			1.251481	1.013245396
1.01-14							3	0.3			0.6138	1.2276			1.236172	0.693434844

Table 6 - D/d=1.01 calculations

Stress Concentration Plot

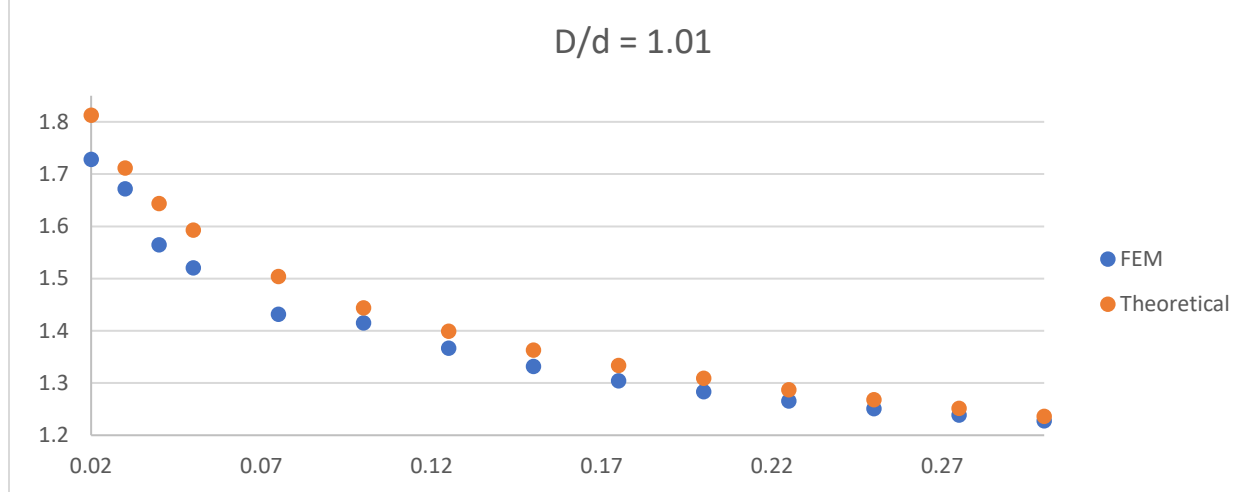


Figure 17 - D/d=1.01 FEM vs Theoretical plot

Results & Summary

The results of both cases turned out to be very similar to the curves from the textbook. The largest percent error between the two was 4.8% and the smallest was 0.7%, with an overall average of about 2.6% error. The general shape of both sets of data very much follow that of the experimental data. Overall, the results of these simulation were very accurate to those from the stress concentration curve.

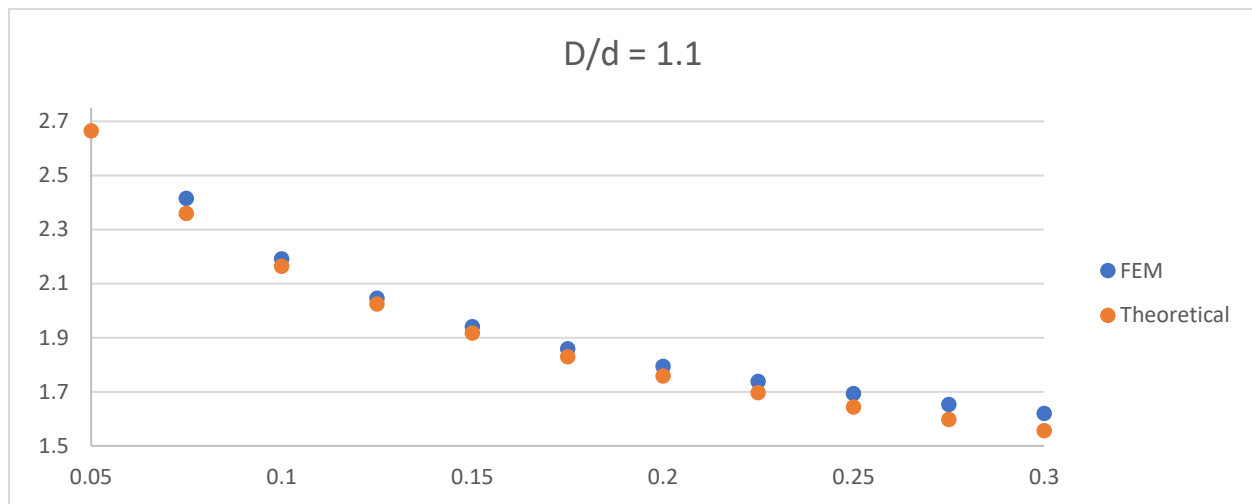


Figure 16 - $D/d=1.1$ FEM vs Theoretical plot

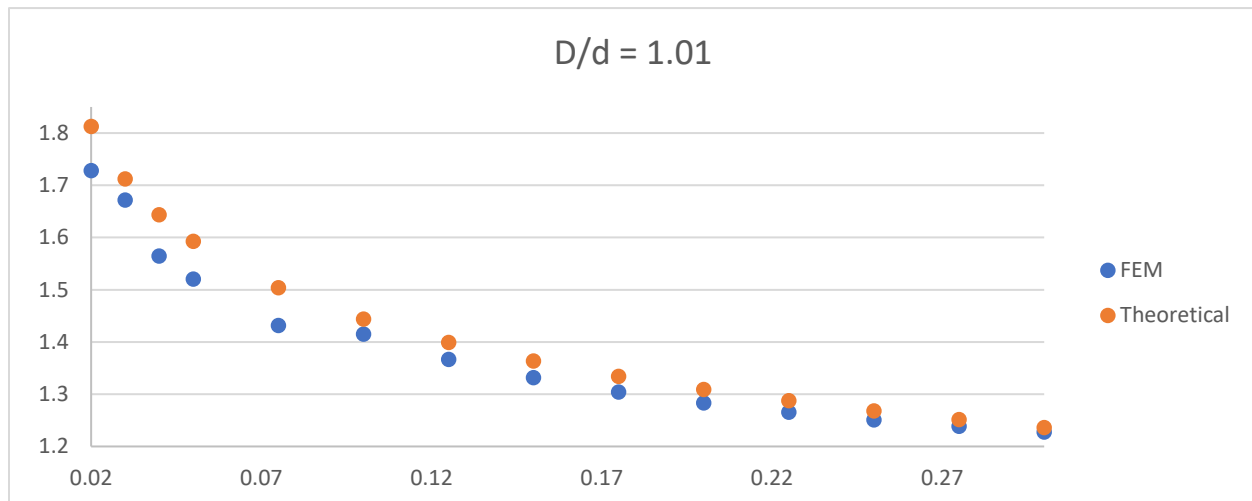


Figure 17 - $D/d=1.01$ FEM vs Theoretical plot

References

Liu, Mingchao, et al. “An Improved Semi-Analytical Solution for Stress at Round-Tip Notches.” *Engineering Fracture Mechanics*, 5 Nov. 2019, hal.science/hal-02310717.